

## AI ASSIGNMENT-6.5

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Batch : 36

### Task Description #1 (AI-Based Code Completion for Conditional Eligibility Check)

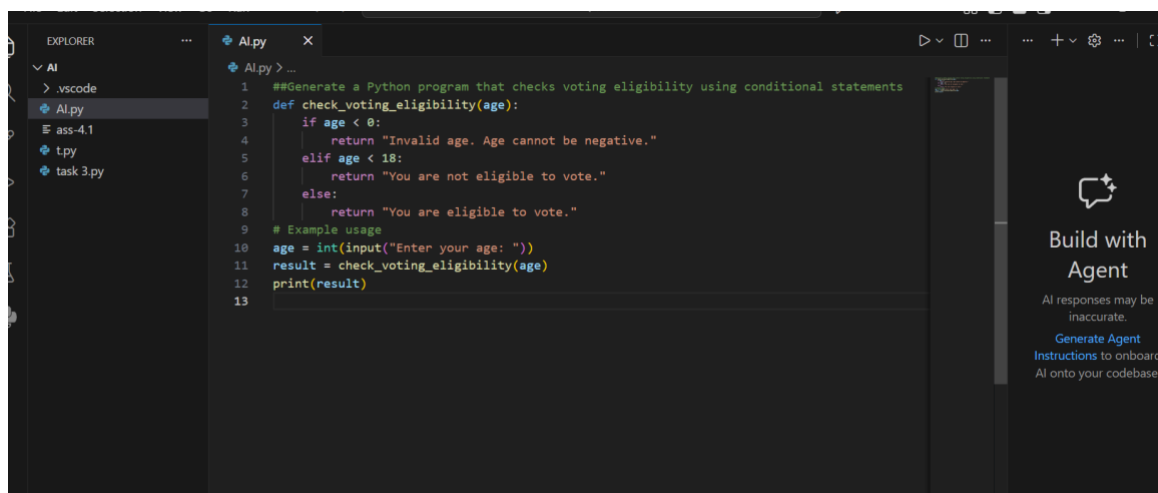
Task: Use an AI tool to generate eligibility logic.

Prompt:

“Generate Python code to check voting eligibility based on age and citizenship.”

Expected Output:

- AI-generated conditional logic.
- Correct eligibility decisions.
- Explanation of conditions.

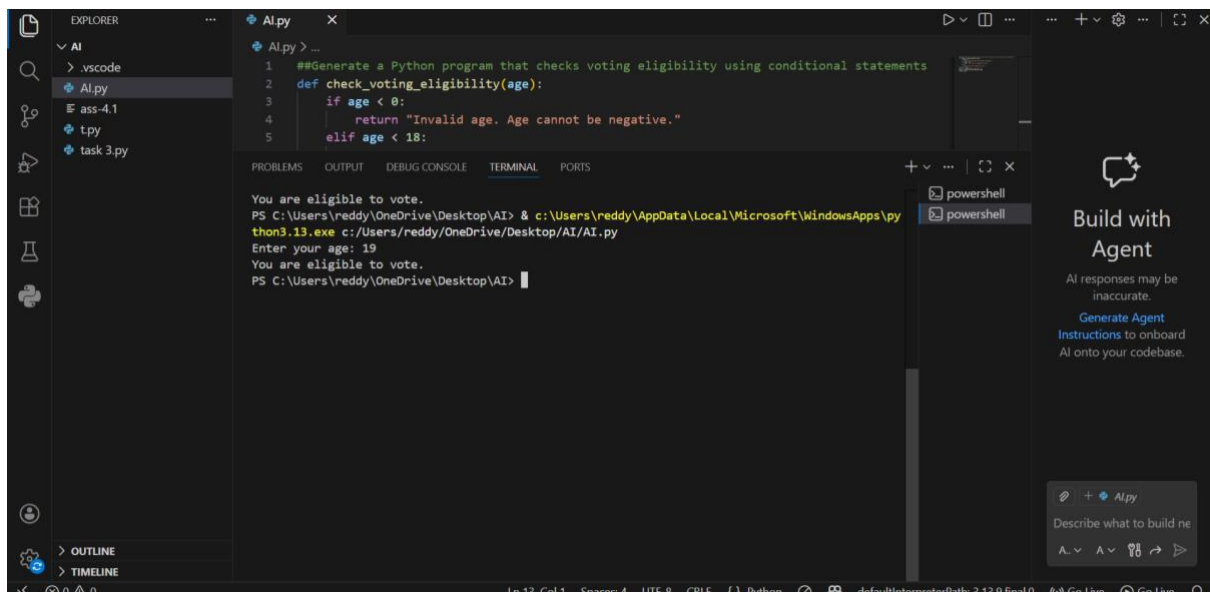


The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'AI' with files 'ass-4.1', 'tpy', and 'task3.py'. The code editor shows a Python file named 'AI.py' with the following code:

```
1  ##Generate a Python program that checks voting eligibility using conditional statements
2  def check_voting_eligibility(age):
3      if age < 0:
4          return "Invalid age. Age cannot be negative."
5      elif age < 18:
6          return "You are not eligible to vote."
7      else:
8          return "You are eligible to vote."
9  # Example usage
10 age = int(input("Enter your age: "))
11 result = check_voting_eligibility(age)
12 print(result)
13
```

On the right side of the code editor, there is a sidebar with the text "Build with Agent" and a small icon of a person with a speech bubble. Below this, it says "AI responses may be inaccurate." and "Generate Agent Instructions to onboard AI onto your codebase."

Output:



## Task Description #2(AI-Based Code Completion for Loop-Based

String Processing)

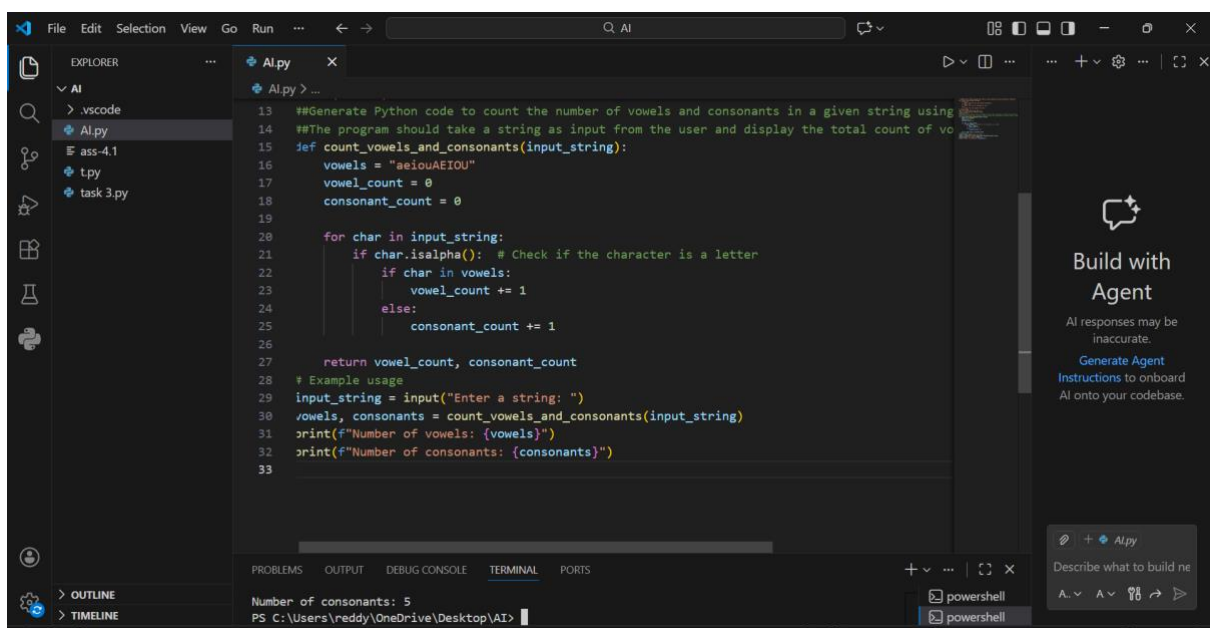
Task: Use an AI tool to process strings using loops.

Prompt:

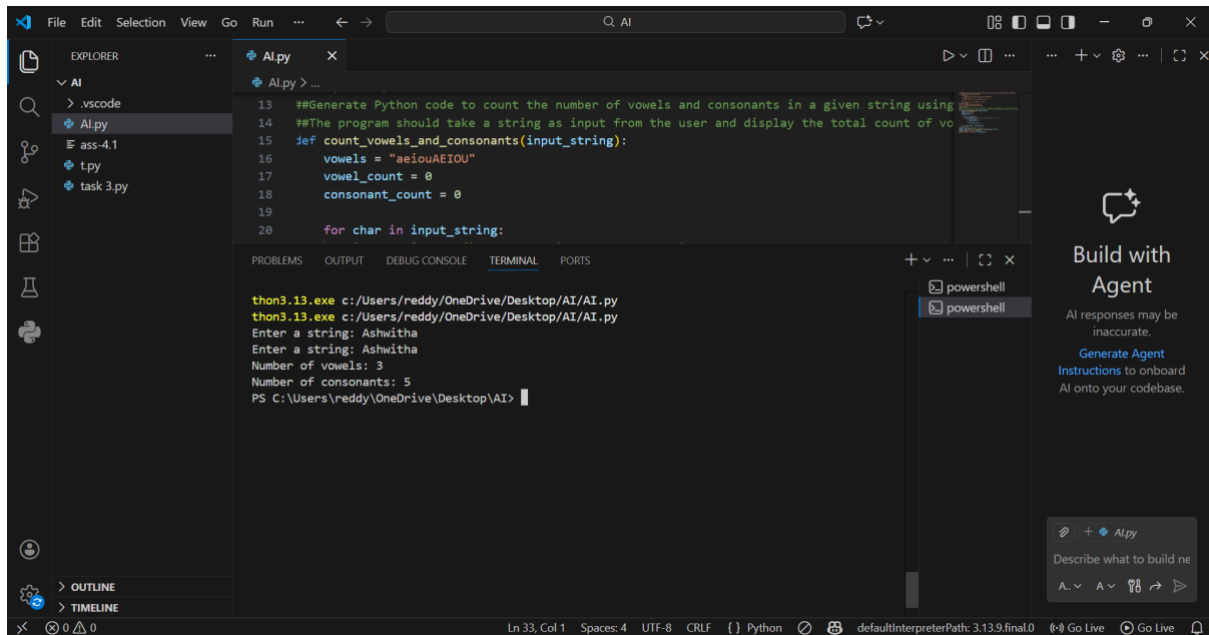
“Generate Python code to count vowels and consonants in a string using a loop.”

Expected Output:

- AI-generated string processing logic.
- Correct counts.
- Output verification.



Output:



The screenshot shows the Visual Studio Code interface. The Explorer panel on the left shows a file named 'AI.py'. The main editor area displays the following Python code:

```
13  ##Generate Python code to count the number of vowels and consonants in a given string using
14  ##The program should take a string as input from the user and display the total count of vo
15  def count_vowels_and_consonants(input_string):
16      vowels = "aeiouAEIOU"
17      vowel_count = 0
18      consonant_count = 0
19
20      for char in input_string:
```

The Output panel at the bottom shows the execution of the script using 'thon3.13.exe'. The output is as follows:

```
thon3.13.exe c:/Users/reddy/OneDrive/Desktop/AI/AI.py
thon3.13.exe c:/Users/reddy/OneDrive/Desktop/AI/AI.py
Enter a string: Ashwitha
Enter a string: Ashwitha
Number of vowels: 3
Number of consonants: 5
PS C:\Users\reddy\OneDrive\Desktop\AI>
```

On the right side of the interface, there is a 'Build with Agent' panel with a message: 'AI responses may be inaccurate. Generate Agent Instructions to onboard AI onto your codebase.'

### Task Description #3 (AI-Assisted Code Completion Reflection

Task)

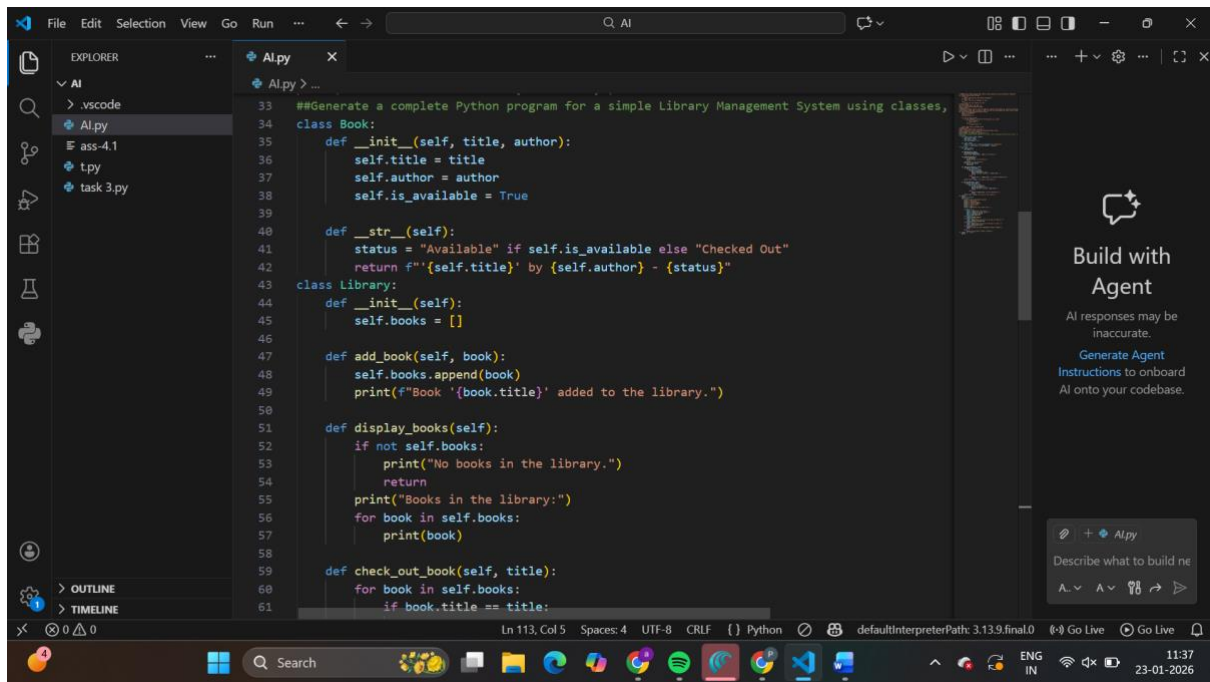
Task: Use an AI tool to generate a complete program using classes, loops, and conditionals.

Prompt:

“Generate a Python program for a library management system using classes, loops, and conditional statements.”

Expected Output:

- Complete AI-generated program.
- Review of AI suggestions quality.
- Short reflection on AI-assisted coding experience.



File Edit Selection View Go Run ... Q AI

EXPLORER

- AI
- .vscode
- Alpy
- ass-4.1
- tpy
- task 3.py

Alpy > ...

```
33 ##Generate a complete Python program for a simple Library Management System using classes,
34 class Book:
35     def __init__(self, title, author):
36         self.title = title
37         self.author = author
38         self.is_available = True
39
40     def __str__(self):
41         status = "Available" if self.is_available else "Checked Out"
42         return f'{self.title}' by {self.author} - {status}"
43 class Library:
44     def __init__(self):
45         self.books = []
46
47     def add_book(self, book):
48         self.books.append(book)
49         print(f"Book '{book.title}' added to the library.")
50
51     def display_books(self):
52         if not self.books:
53             print("No books in the library.")
54             return
55         print("Books in the library:")
56         for book in self.books:
57             print(book)
58
59     def check_out_book(self, title):
60         for book in self.books:
61             if book.title == title:
```

Ln 113, Col 5 Spaces: 4 UTF-8 CRLF {} Python defaultInterpreterPath: 3.13.9.final.0 Go Live Go Live

Search

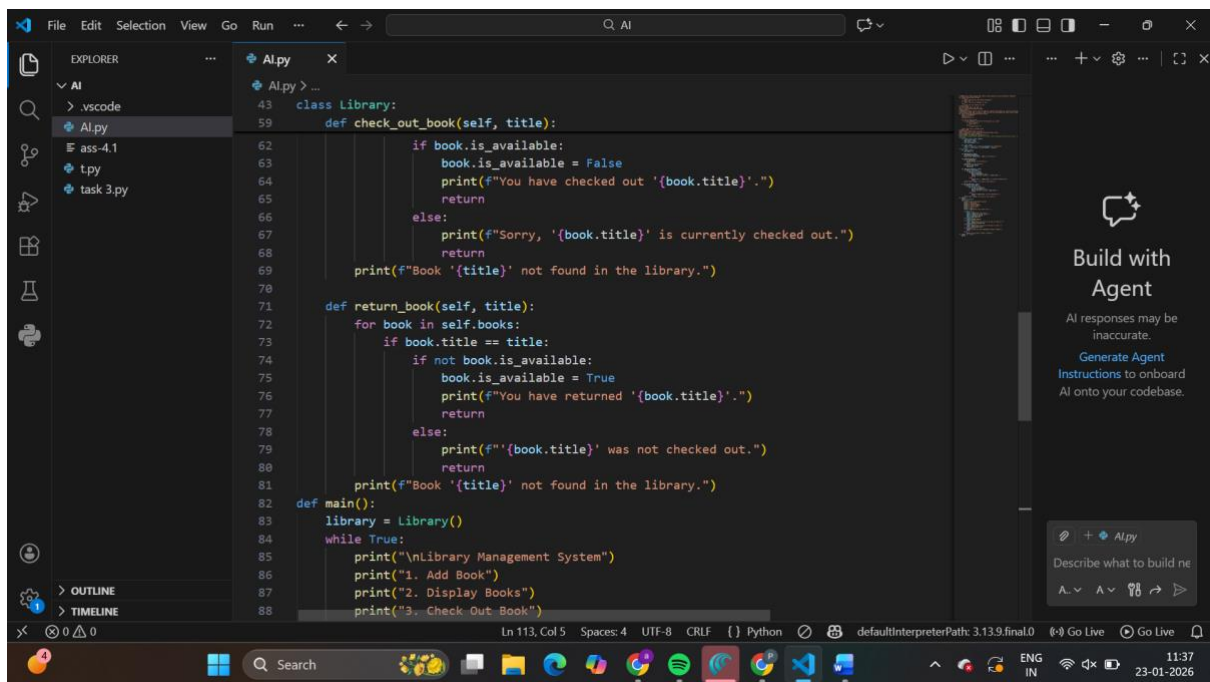
Build with Agent

AI responses may be inaccurate.

Generate Agent

Instructions to onboard AI onto your codebase.

Describe what to build ne



File Edit Selection View Go Run ... Q AI

EXPLORER

- AI
- .vscode
- Alpy
- ass-4.1
- tpy
- task 3.py

Alpy > ...

```
43 class Library:
59     def check_out_book(self, title):
62         if book.is_available:
63             book.is_available = False
64             print(f"You have checked out '{book.title}'")
65             return
66         else:
67             print(f"Sorry, '{book.title}' is currently checked out.")
68             return
69         print(f"Book '{title}' not found in the library.")
70
71     def return_book(self, title):
72         for book in self.books:
73             if book.title == title:
74                 if not book.is_available:
75                     book.is_available = True
76                     print(f"You have returned '{book.title}'")
77                     return
78                 else:
79                     print(f"'{book.title}' was not checked out.")
80                     return
81             print(f"Book '{title}' not found in the library.")
82
83 def main():
84     library = Library()
85     while True:
86         print("\nLibrary Management System")
87         print("1. Add Book")
88         print("2. Display Books")
89         print("3. Check Out Book")
```

Ln 113, Col 5 Spaces: 4 UTF-8 CRLF {} Python defaultInterpreterPath: 3.13.9.final.0 Go Live Go Live

Search

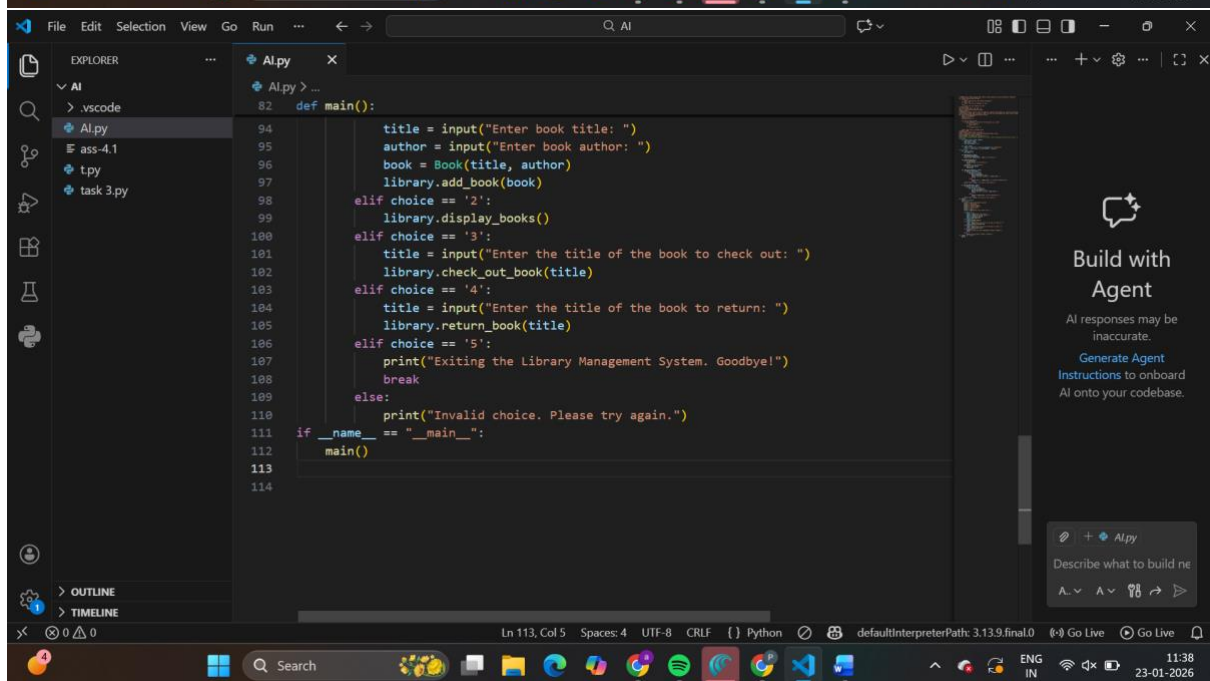
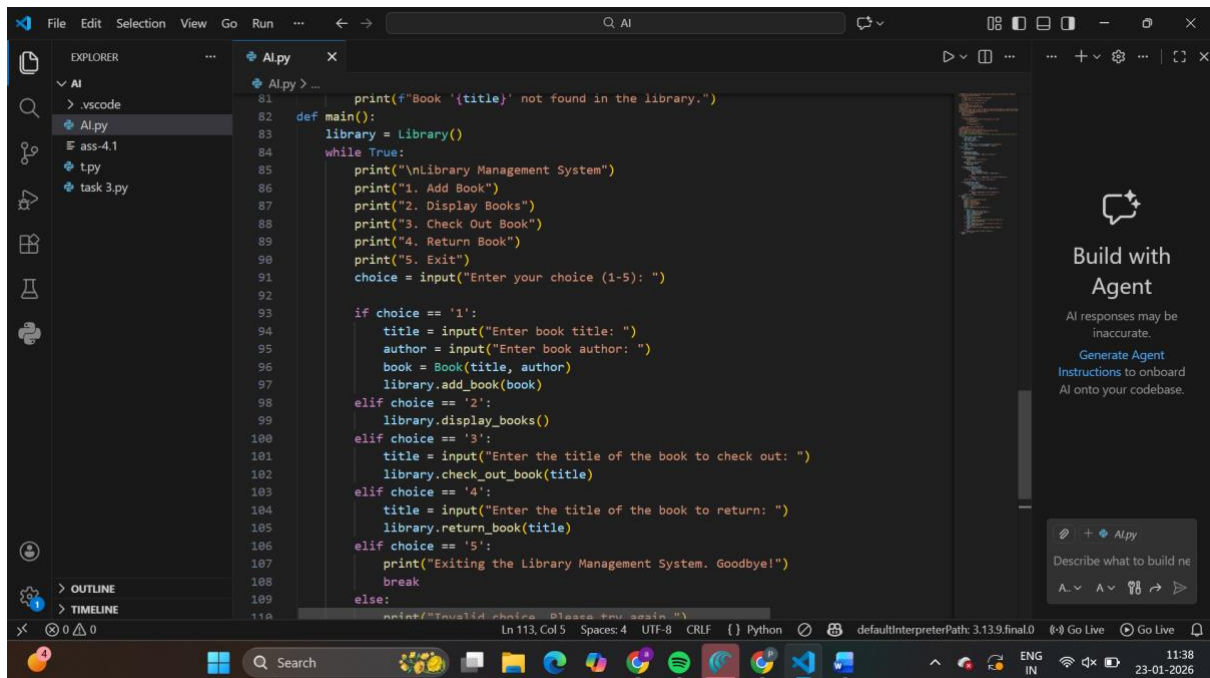
Build with Agent

AI responses may be inaccurate.

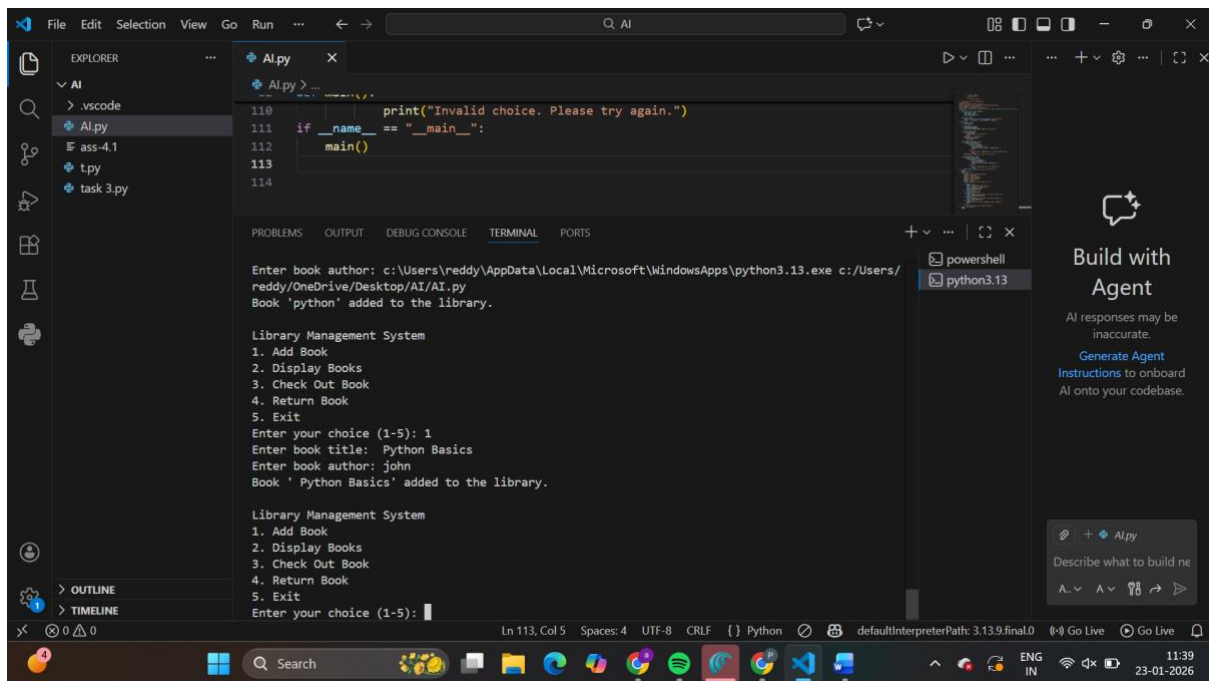
Generate Agent

Instructions to onboard AI onto your codebase.

Describe what to build ne



Output:



The screenshot shows a Visual Studio Code editor window with a Python file named `Al.py` open. The code in the editor is as follows:

```
110         print("Invalid choice. Please try again.")
111     if __name__ == "__main__":
112         main()
113
114
```

The terminal window at the bottom shows the execution of the script. It prompts the user to enter a book author, then displays a menu for the Library Management System. The user enters '1' for 'Add Book', then enters 'Python Basics' as the book title and 'john' as the author. The script confirms the book has been added to the library and prompts the user to enter a choice (1-5). The user enters '1'.

```
Enter book author: c:\Users\reddy\AppData\Local\Microsoft\WindowsApps\python3.13.exe c:/Users/
reddy/OneDrive/Desktop/AI/AI.py
Book 'python' added to the library.

Library Management System
1. Add Book
2. Display Books
3. Check Out Book
4. Return Book
5. Exit
Enter your choice (1-5): 1
Enter book title: Python Basics
Enter book author: john
Book ' Python Basics' added to the library.

Library Management System
1. Add Book
2. Display Books
3. Check Out Book
4. Return Book
5. Exit
Enter your choice (1-5):
```

The status bar at the bottom indicates the file is at line 113, column 5, using UTF-8 encoding with CRLF line endings. The default interpreter path is `3.13.9.final.0`.

#### Task Description #4 (AI-Assisted Code Completion for Class-Based Attendance System)

Task: Use an AI tool to generate an attendance management class.

Prompt: “Generate a Python class to mark and display student attendance using loops.”

Expected Output:

- AI-generated attendance logic.
- Correct display of attendance.



File Edit Selection View Go Run ... < -> Q AI

EXPLORER

AI

.vscode

Al.py

ass-4.1

tpy

task 3.py

OUTLINE

TIMELINE

Al.py

Al.py > ...

113 ##Generate a Python class to mark and display student attendance using loops.

114 ##The program should allow marking students as present or absent and display the attendance

115 class Student:

116 def \_\_init\_\_(self, name):

117 self.name = name

118 self.attendance = []

119

120 def mark\_present(self):

121 self.attendance.append("Present")

122

123 def mark\_absent(self):

124 self.attendance.append("Absent")

125

126 def display\_attendance(self):

127 print(f"Attendance for {self.name}:")

128 for i, status in enumerate(self.attendance, start=1):

129 print(f"Day {i}: {status}")

130

131

132 class AttendanceSystem:

133 def \_\_init\_\_(self):

134 self.students = []

135

136 def add\_student(self, student):

137 self.students.append(student)

138

139 def display\_all\_attendance(self):

140 for student in self.students:

141 student.display\_attendance()

142

143

144

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Ln 179, Col 1 Spaces: 4 UTF-8 CRLF {} Python defaultInterpreterPath: 3.13.9.final.0 Go Live Go Live

Build with Agent

AI responses may be inaccurate.

Generate Agent

Instructions to onboard AI onto your codebase.

Describe what to build ne

File Edit Selection View Go Run ... < -> Q AI

EXPLORER

AI

.vscode

Al.py

ass-4.1

tpy

task 3.py

OUTLINE

TIMELINE

Al.py

Al.py > ...

143

144

145 def main():

146 attendance\_system = AttendanceSystem()

147

148 # Input validation

149 num\_students = int(input("Enter the number of students: "))

150

151 for \_ in range(num\_students):

152 name = input("Enter student name: ")

153 student = Student(name)

154 attendance\_system.add\_student(student)

155

156 num\_days = int(input("Enter the number of days to mark attendance: "))

157

158 for day in range(1, num\_days + 1):

159 print(f"\nMarking attendance for Day {day}")

160 for student in attendance\_system.students:

161 while True:

162 status = input(f"Is {student.name} present? (y/n): ").strip().lower()

163 if status == 'y':

164 student.mark\_present()

165 break

166 elif status == 'n':

167 student.mark\_absent()

168 break

169 else:

170 print("Invalid input! Please enter 'y' or 'n'.")

171

Ln 179, Col 1 Spaces: 4 UTF-8 CRLF {} Python defaultInterpreterPath: 3.13.9.final.0 Go Live Go Live

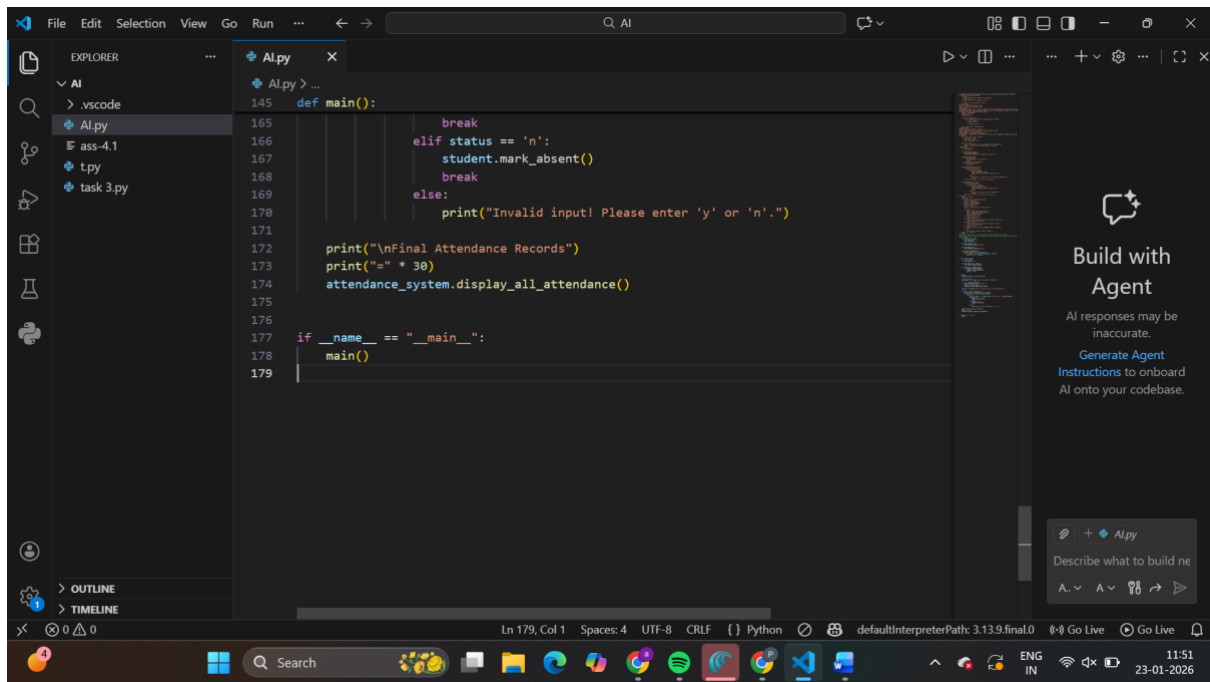
Build with Agent

AI responses may be inaccurate.

Generate Agent

Instructions to onboard AI onto your codebase.

Describe what to build ne



Output:



File Edit Selection View Go Run ... AI

EXPLORER

AI  
  .vscod  
    Al.py  
    ass-4.1  
    tpy  
    task 3.py

Al.py

145 def main():  
PS C:\Users\reddy\OneDrive\Desktop\AI> & c:\Users\reddy\AppData\Local\Microsoft\WindowsApps\python3.13.exe c:/Users/redd...  
sers/reddy/OneDrive/Desktop/AI/AI.py  
Enter the number of students: 4  
Enter student name: Ashwutha  
Enter student name: Harshini  
Enter student name: Akshitha  
Enter student name: varshitha  
Enter the number of days to mark attendance: 4  
  
Marking attendance for Day 1  
Is Ashwutha present? (y/n): y  
Is Harshini present? (y/n): y  
Is Akshitha present? (y/n): y  
Is varshitha present? (y/n): y  
  
Marking attendance for Day 2  
Is Ashwutha present? (y/n): n  
Is Harshini present? (y/n): y  
Is Akshitha present? (y/n): y  
Is varshitha present? (y/n): y  
  
Marking attendance for Day 3  
Is Ashwutha present? (y/n): y  
Is Harshini present? (y/n): n  
Is Akshitha present? (y/n): n  
Is varshitha present? (y/n): y  
  
Marking attendance for Day 4  
Is Ashwutha present? (y/n): y

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

powerShell

Build with Agent  
AI responses may be inaccurate.  
Generate Agent  
Instructions to onboard AI onto your codebase.  
+ Al.py  
Describe what to build ne  
A... A... ? ? ? ?

Ln 179, Col 1 Spaces: 4 UTF-8 CRLF {} Python defaultInterpreterPath: 3.13.9.final.0 Go Live Go Live

File Edit Selection View Go Run ... AI

EXPLORER

AI  
  .vscod  
    Al.py  
    ass-4.1  
    tpy  
    task 3.py

Al.py

145 def main():  
Is Akshitha present? (y/n): n  
Is varshitha present? (y/n): y  
  
Marking attendance for Day 4  
Is Ashwutha present? (y/n): y  
Is Harshini present? (y/n): y  
Is Akshitha present? (y/n): y  
Is varshitha present? (y/n): y  
  
Final Attendance Records  
=====

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

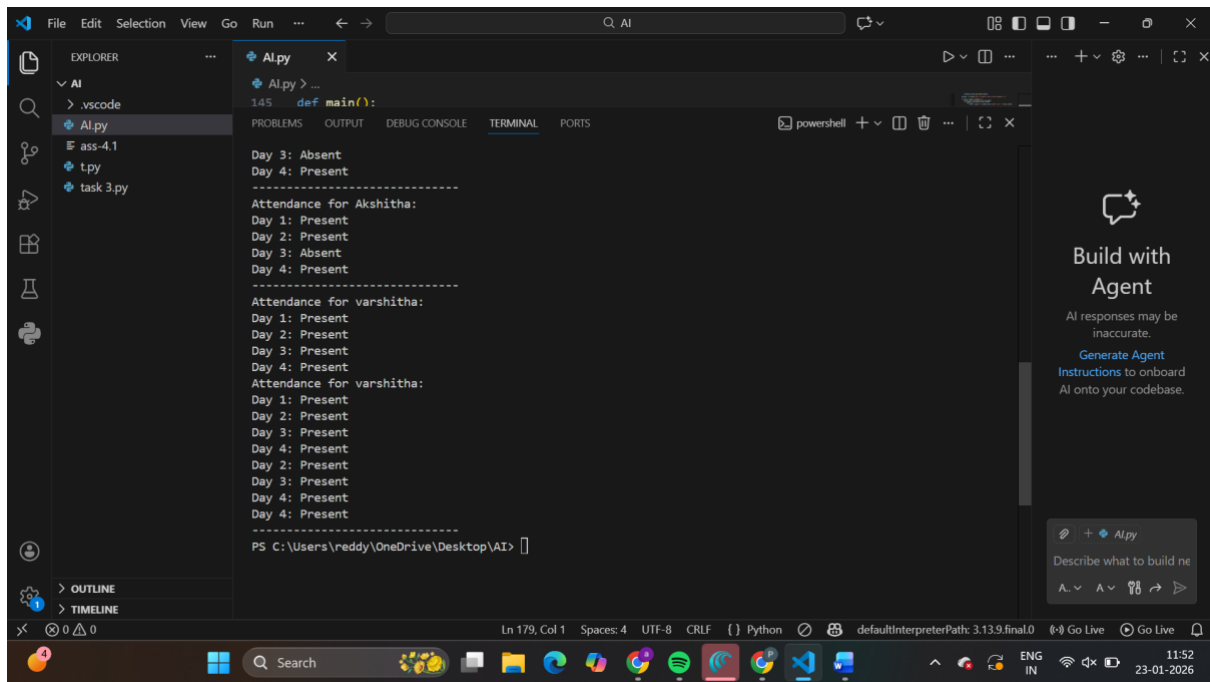
PORTS

powerShell

Build with Agent  
AI responses may be inaccurate.  
Generate Agent  
Instructions to onboard AI onto your codebase.  
+ Al.py  
Describe what to build ne  
A... A... ? ? ? ?

Ln 179, Col 1 Spaces: 4 UTF-8 CRLF {} Python defaultInterpreterPath: 3.13.9.final.0 Go Live Go Live

Attendance for Ashwutha:  
Day 1: Present  
Day 2: Absent  
Day 3: Present  
Day 4: Present  
-----  
Attendance for Harshini:  
Day 1: Present  
Day 2: Present  
Day 3: Absent  
Day 4: Present  
-----  
Attendance for Akshitha:  
Day 1: Present  
Day 2: Present  
Day 3: Absent  
Day 4: Present  
-----



## Task Description #5 (AI-Based Code Completion for Conditional

Menu Navigation)

Task: Use an AI tool to complete a navigation menu.

Prompt: “Generate a Python program using loops and conditionals to simulate an ATM menu.”

Expected Output:

- AI-generated menu logic.
- Correct option handling.
- Output verification.

The screenshot shows the Visual Studio Code editor with a Python file named `Al.py`. The code defines an `atm_menu()` function that initializes a balance of 1000 and enters a loop where it displays an ATM menu. The menu options are: 1. Check Balance, 2. Deposit, 3. Withdraw, and 4. Exit. The code handles the '1' option by displaying the current balance and the '2' option by depositing a specified amount. It also includes error handling for invalid choices and non-positive deposit amounts. The Explorer sidebar on the left shows the project structure with files `ass-4.1`, `tpy`, and `task 3.py`. The bottom status bar indicates the file is at line 222, column 5, using UTF-8 encoding with CRLF line endings.

```
185 #Withdraw
186 #Exit
187 def atm_menu():
188     balance = 1000 # Initial balance
189
190     while True:
191         print("\nATM Menu:")
192         print("1. Check Balance")
193         print("2. Deposit")
194         print("3. Withdraw")
195         print("4. Exit")
196
197         choice = input("Please select an option (1-4): ")
198
199         if choice == '1':
200             print(f"Your current balance is: ${balance:.2f}")
201         elif choice == '2':
202             amount = float(input("Enter amount to deposit: $"))
203             if amount > 0:
204                 balance += amount
205                 print(f"${amount:.2f} deposited successfully.")
206             else:
207                 print("Invalid amount. Please enter a positive value.")
208         elif choice == '3':
209             amount = float(input("Enter amount to withdraw: $"))
210             if 0 < amount <= balance:
211                 balance -= amount
212                 print(f"${amount:.2f} withdrawn successfully.")
213             else:
```

This screenshot shows the same VS Code editor with the `Al.py` file, but now the `Terminal` panel is active at the bottom. It displays the output of running the program. The user has entered '1' to check the balance, which is shown as \$1000.00. Then, the user entered '2' to deposit, and entered '1000' as the amount, resulting in a successful deposit message. The code in the editor is now at line 222, column 5. The status bar shows the interpreter path as `python3.13`.

```
187 def atm_menu():
219     print("Invalid choice. Please select a valid option.")
220 if __name__ == "__main__":
221     atm_menu()
222
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Please select an option (1-4): 1  
Your current balance is: \$1000.00

ATM Menu:  
1. Check Balance  
2. Deposit  
Your current balance is: \$1000.00

ATM Menu:  
1. Check Balance  
2. Deposit  
3. Withdraw  
1. Check Balance  
2. Deposit  
3. Withdraw  
4. Exit  
Please select an option (1-4): 2  
Enter amount to deposit: \$1000  
\$1000.00 deposited successfully.

